

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence and Application Development
PRACTICAL IV

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Class: TYIT	Batch:2
Date of Assignment: 18-07-26	Date/Time of Submission: 18-07-26

A] Develop a simple JSP application to display values obtained from these of intrinsic objects of various types. CODE:-

index.jsp:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<html><head><title> JSP Page</title></head>
<body>
<h1>Use of Intrinsic Objects in JSP</h1>
<h1>Request Object </h1>
Query String <%=request.getQueryString() %><br>
Context Path <%=request.getContextPath() %><br>
Remote Host <%=request.getRemoteHost() %><br>
<h1>Response Object </h1>
Character Encoding Type <%=response.getCharacterEncoding() %><br>
Content Type
<%=response.getContentType() %><br>
Locale <%=response.getLocale() %><br>
<h1>Session Object </h1>
ID <%=session.getId() %><br>
Creation Time <%=new java.util.Date(session.getCreationTime()) %><br>
Last Access Time<%=new java.util.Date(session.getLastAccessedTime()) %><br>
</body>
</html>
```

OUTPUT:-

Use of Intrinsic Objects in JSP

Request Object

```
Query String null
Context Path /Prac4
Remote Host 0:0:0:0:0:0:1
```

Response Object

```
Character Encoding Type UTF-8
Content Type text/html;charset=UTF-8
Locale en_IN
```

Session Object

```
ID 45fed818ce2c22cabbefb59c2c79
Creation Time Sat Jul 18 14:07:51 IST 2026
Last Access Time Sat Jul 18 14:07:51 IST 2026
```

khushii

B] Develop a simple JSP application to pass values from one page to another with validations. (Name-txt, age txt, hobbies-checkbox, email-txt, gender-radio button).

CODE:-

index.jsp:

```
<!DOCTYPE html>
<html>
<head>
<meta https-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>JSP Registration</title>
</head>
<body>
<h1>Registration Form</h1>
<form action="val.jsp" method="post">
<table style="width:50%">
<tr>
<td>Full Name</td>
<td><input type="text" name="fullname"/></td>
</tr><tr>
<td>Age</td>
<td><input type="text" name="age"/></td>
</tr><tr>
<td>E-Mail</td>
<td><input type="text" name="email" size="20"/></td>
</tr><tr>
<td>Gender</td>
<td><input type="radio" name="gender" value="Male">Male
<input type="radio" name="gender" value="Female">Female
</td>
</tr><tr>
<td>Hobbies</td>
<td><input type="checkbox" name="hb" value="Acting"/>Acting
<input type="checkbox" name="hb" value="Dancing"/>Dancing
<input type="checkbox" name="hb" value="Singing"/>Singing
<input type="checkbox" name="hb" value="Drawing"/>Drawing
</td>
</tr>
</table>
<input type="submit" value="Register"/>
</form>
</body>
</html>
```

val.jsp:

```
<%@page language="java" contentType="text/html" pageEncoding="UTF-8" %>
<%!
int ageInNumbers;
private static final String EMAIL_REGEX="^[\\w-\\+]+(\\.\\[\\w\\+])*@[\\w-]+(\\.\\[\\w\\+])*(\\.[a-
z]{2,})$"; %>
<%
String name=request.getParameter("fullname");
String age=request.getParameter("age");
String email=request.getParameter("email");
String gender=request.getParameter("gender");
String hb[]=request.getParameterValues("hb");
if (name == null || name.isEmpty() || age == null || age.isEmpty() || email == null || email.isEmpty() || gender ==
```

```

    null || gender.isEmpty()) {
out.println("<font color=red>Please fill all the fields</font><br>");
    }
    if(!email.matches(EMAIL_REGEX)){
out.println("<font color=red>Correct Your Email Address</font><br>");
    }
    try{
ageInNumbers=Integer.parseInt(age.trim());
    }
    catch(NumberFormatException e){
out.println("<font color=red>Age must be numbers</font><br>");
    }
    if (ageInNumbers < 18 || ageInNumbers > 60) {
out.println("<font color=red>Age must be between 18 and 60</font><br>");
    }
    %>
Your Extended Information is as follows:<br><br>
Full Name<b>:<%=name%></b><br>
Age<b>:<%=age%></b><br>
EMail<b>:<%=email%></b><br>
Gender<b>:<%=gender%></b><br>
Hobbies<b>:
<%
if(hb!=null && hb.length!=0){
for(int i=0;i<hb.length;i++){
out.println(hb[i]);
}
}
%>

```

web.xml:

```

<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org/xml/ns/javaee/web
app_3_1.xsd"
version="3.1">
<welcome-file-list>
<welcome-file>index.jsp</welcome-file>
</welcome-file-list>
</web-app>

```

OUTPUT:-

Registration Form

Full Name	KHUSHI
Age	20
E-Mail	khushipatelmvlu@gmail.com
Gender	☐ Male ☒ Female
Hobbies	☐ Acting ☒ Dancing ☐ Singing ☐ Drawing
Register	

Your Extended Information is as follows:

Full Name:**KHUSHI**
 Age:**20**
 EMail:**khushipatelmvlu@gmail.com**
 Gender:**Female**
 Hobbies: **Dancing**

C] Create a registration and login JSP application to register and authentication the user based on username & password using JDC .

CODE:-

```

index.jsp
<!DOCTYPE html>
<html>
<head>
<title>Home</title>
</head>
<body>
<h2>Registration & Login System</h2>
<a href="register.jsp">Register</a>
<br><br>
<a href="login.jsp">Login</a>
</body>
</html>
register.jsp:
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<%@ page import="db.DBConnection" %>
<%
String msg = "";
if(request.getMethod().equalsIgnoreCase("POST"))
{
    String username = request.getParameter("username");
    String password = request.getParameter("password");
    Connection con = DBConnection.getConnection();
    if(con == null)
    {
        msg = "Database Connection Failed";
    }
    else
    {
        try
        {
            PreparedStatement check = con.prepareStatement(
                "SELECT * FROM users WHERE username=?");
            check.setString(1, username);
            ResultSet rs = check.executeQuery();
            if(rs.next())
            {
                msg = "Username Already Exists";
            }
            else
            {
                PreparedStatement ps = con.prepareStatement(
                    "INSERT INTO users(username,password) VALUES(?,?)");
                ps.setString(1, username);
                ps.setString(2, password);
                int i = ps.executeUpdate();
                if(i > 0)
                {
                    msg = "Registration Successful";
                }
            }
            else
            {

```

```

        msg = "Registration Failed";
    }

    ps.close();
    rs.close();
    check.close();
    con.close();
}
catch(Exception e)
{
    msg = e.getMessage();
} } }

%>
<!DOCTYPE html>
<html>
<head>
    <title>User Registration</title>
</head>
<body>
<h2>User Registration</h2>
<form method="post">
<table>
<tr>
<td>Username</td>
<td><input type="text" name="username" required></td>
</tr>
<tr>
<td>Password</td>
<td><input type="password" name="password" required></td>
</tr>
<tr>
<td></td>
<td><input type="submit" value="Register"></td>
</tr>
</table>
</form>
<br>
<font color="red">
<%= msg %>
</font>
<br><br>
<a href="login.jsp">Go to Login</a>
</body>
</html>
login.jsp
<%@ page import="java.sql.*" %>
<%@ page import="db.DBConnection" %>
<%
String msg = "";
if(request.getMethod().equalsIgnoreCase("POST"))
{
    String username = request.getParameter("username");
    String password = request.getParameter("password");
    Connection con = DBConnection.getConnection();
    PreparedStatement ps = con.prepareStatement(
        "SELECT * FROM users WHERE username=? AND password=?");
    ps.setString(1, username);
    ps.setString(2, password);

```

```

ResultSet rs = ps.executeQuery();
if(rs.next())
{
    session.setAttribute("username", username);
    response.sendRedirect("success.jsp");
}
else
{
    msg = "Invalid Username or Password";
}
}
%>
<!DOCTYPE html>
<html>
<head>
<title>Login</title>
</head>
<body>
<h2>Login</h2>
<form method="post">
Username :
<input type="text" name="username" required>
<br><br>
Password :
<input type="password" name="password" required>
<br><br>
<input type="submit" value="Login">
</form>
<br>
<%=msg%>
</body>
</html>
success.jsp
<%
String user = (String)session.getAttribute("username");
if(user == null)
{
    response.sendRedirect("login.jsp");
    return;
}
%>
<!DOCTYPE html>
<html>
<head>
<title>Success</title>
</head>
<body>
<h2>Login Successful</h2>
<h3>Welcome <%=user%></h3>
<a href="index.jsp">Logout</a>
</body>
</html>
DBConnection.java:
package db;
import java.sql.Connection;
import java.sql.DriverManager;
public class DBConnection {
    public static Connection getConnection() {

```

```

Connection con = null;
try {
    Class.forName("com.mysql.cj.jdbc.Driver");
    String url = "jdbc:mysql://localhost:3306/db?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";
    con = DriverManager.getConnection(url, "root", "khup@0103");
    System.out.println("Database Connected Successfully");
} catch (Exception e) {
    e.printStackTrace();
}
return con;
}
}

```

SQL :-

```
CREATE DATABASE userdb;
```

```
USE userdb;
```

```
CREATE TABLE users
```

```

(
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(50) UNIQUE,
    password VARCHAR(50)
);

```

OUTPUT:-**User Registration**

Username

Password

Registration Successful

[Go to Login](#)

Login

Username :

Password : 

Login Successful

Welcome khushii

[Logout](#)

